

DAVID LIU

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EDUCATION

University of Michigan | College of Engineering
Bachelor's in Computer Science, Minor in Mathematics

Ann Arbor, Michigan
Aug 2021 – May 2025

- **GPA:** 3.774
- **Programming Languages:** Python, C/C++, JavaScript, React.js, Node.js, Golang
- **Relevant Coursework:** Web Systems, Operating Systems, Computer Architecture, Computer Networks

WORK EXPERIENCE

Whova

San Diego, CA

Software Engineer

July 2025 – Present

- Led full-stack development for Whova registration features, including payment flows and third-party integrations
- Implemented and maintained OAuth 2.0-based authorization flows for external service integrations
- Configured and monitored Datadog dashboards and alerts to improve system reliability, and incident response
- Collaborated with product managers and designers to scope requirements, evaluate tradeoffs, and drive technical decisions from design to production

Whova

San Diego, CA

Software Engineer Intern

May 2024 – Aug 2024

- Created and integrated an Agenda Conflict Checker for the Whova EMS Dashboard in an Agile environment
- Adapted backend logic and data workflows to evolving product requirements while maintaining system reliability
- Analyzed and presented feature outcomes to stakeholders, iterating on user experience based on feedback
- Enhanced security with Datadog monitoring, and added unit/integration testing for project release

EXTRACURRICULARS

Tech For Social Good

Ann Arbor, MI

Lead Engineer

March 2024 – May 2025

- Worked with nonprofit clients to gather requirements and design technical solutions aligned with organizational goals
- Led engineering efforts by translating client needs into clear technical tasks and system designs
- Coordinated with cross-functional team members to deliver features on schedule and adapt to evolving project needs
- Reviewed code and mentored team members to maintain code quality and engineering best practices

University of Michigan

Ann Arbor, MI

Assistant Instructor

March 2023 – May 2025

- Taught Computer Architecture classes and guided students through associated lab work
- Planned study sessions and prepared lectures that would review content in the respective classes before exams
- Hosted one-on-one tutoring sessions that would answer questions about the homework and course content
- Collaborated with graduate student instructors to hold group office hours for students in respective classes

PROJECTS

Movie Match (Node.js, React.js, MongoDB, Express.js, Tailwind CSS)

Personal Project

- Social media web app that uses letterbox APIs to match users by movie/show preferences using OAuth 2.0
- Users can sign in and sign out of their profiles using their Google account using OpenID connect
- Built frontend using React and Redux for state management, backend with Node/Express and MongoDB

Material Structure Generator (Python, Pandas, Pytorch, Tensorflow)

Personal Project

- Clustered material database based on shared qualities to create training sets for an adaptive domain ML model
- Given an inputted chemical formula, outputs structures with lowest bond energy formation
- Implemented scalable solution for processing large datasets, improving the efficiency of material structure predictions

Operating System (C++)

Class Project

- Implemented a mock operating system with a thread library, networked file system, and a pager
- Designed and optimized data structures for efficient thread scheduling, file access, and virtual memory management.